

Introduction

Antibiotic residues have been leaching into wastewater, causing the bacteria to develop **Antibiotic Resistance Genes** or ARG's in order to stay alive. The development of ARG's in the bacteria can render antibiotics a useless treatment against sickness.

The introduction of algae, specifically ***Chlorella pyrenoidosa***, has been hypothesized to act as a cleaner, easier method to remove the antibiotic residues from water. To deduce how this method would impact the organisms in the water, ***Daphnia magna*** were used to measure the effects of this method. *Daphnia magna* is a small macroinvertebrate commonly used for ecotoxicological testing due to its role in the food chain and its susceptibility to pollutants.

Can algae be used as an **environmentally friendly water treatment**, to reduce the amount of antibiotic residues in water, without heavily affecting the health of organisms residing in that water?



Image of Antibiotic magazine from <https://www.carolina.com/microbiology-supplies/economy-antibiotic-disk-tetracycline-30-mcg/806500.pr>

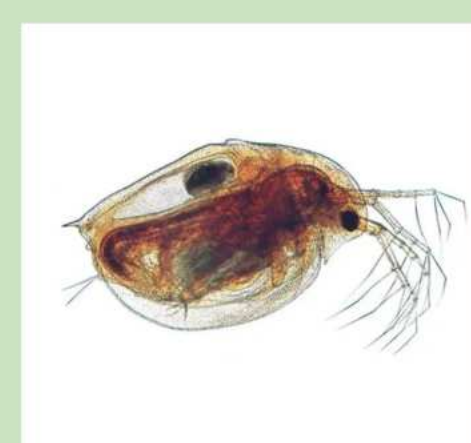


Image of *D. magna* from <https://www.homesciencetools.com/product/mixed-crustacean-dry-eggs/>



Image of *C. pyrenoidosa* from https://algaeresearchsupply.com/products/algae-research-supply-algae-culture-kit-for-chlorella-pyrenoidosa?srsltid=AfmBOpuwSFZOC18_dgl2_Ay6EAdVvuyLwZD-6KDJyKgLKn463ux4

Objectives

Purpose: To determine the effects algae on Erythromycin and Ampicillin and the algae's effect on *Daphnia magna*.

Hypothesis: If algae is added into the wastewater containing trace amounts of antibiotics with *D. magna* then it will adsorb the smaller molecules of the antibiotics and negatively affect the *D. magna*.

Independent Variable: The concentrations of the two types of antibiotics being tested.

Dependent Variable: The health of the algae being tested and the health of the *D. magna*.

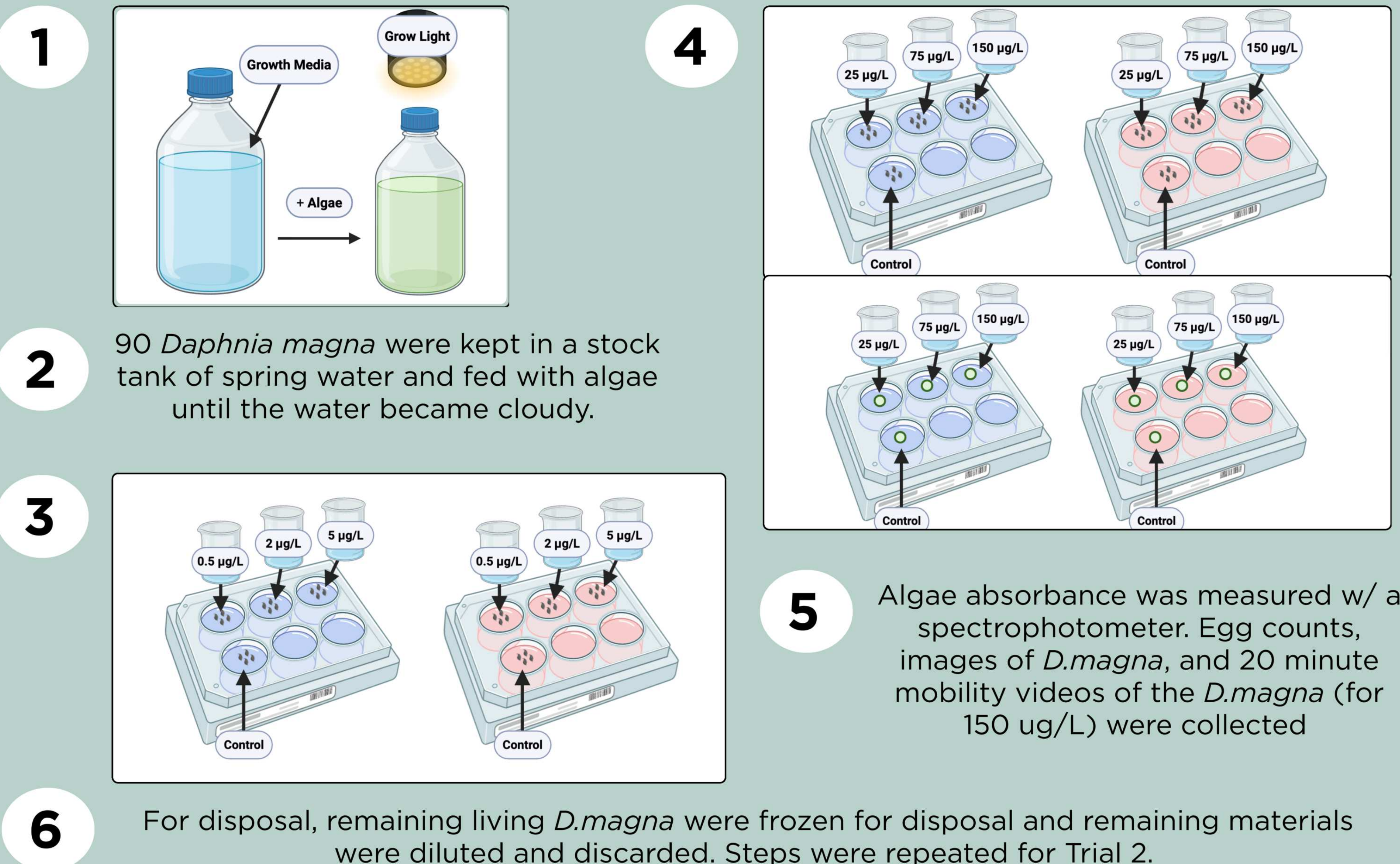
Constants: The type of algae being tested, *Chlorella pyrenoidosa*, and the types of antibiotics being tested.

Controls:

- Well containing spring water and *Chlorella pyrenoidosa*.
- Well containing spring water, *Chlorella pyrenoidosa*, and *D. magna*.

Determining the Effect of Erythromycin and Ampicillin on *Chlorella pyrenoidosa* and *Daphnia magna*

Methods



Results

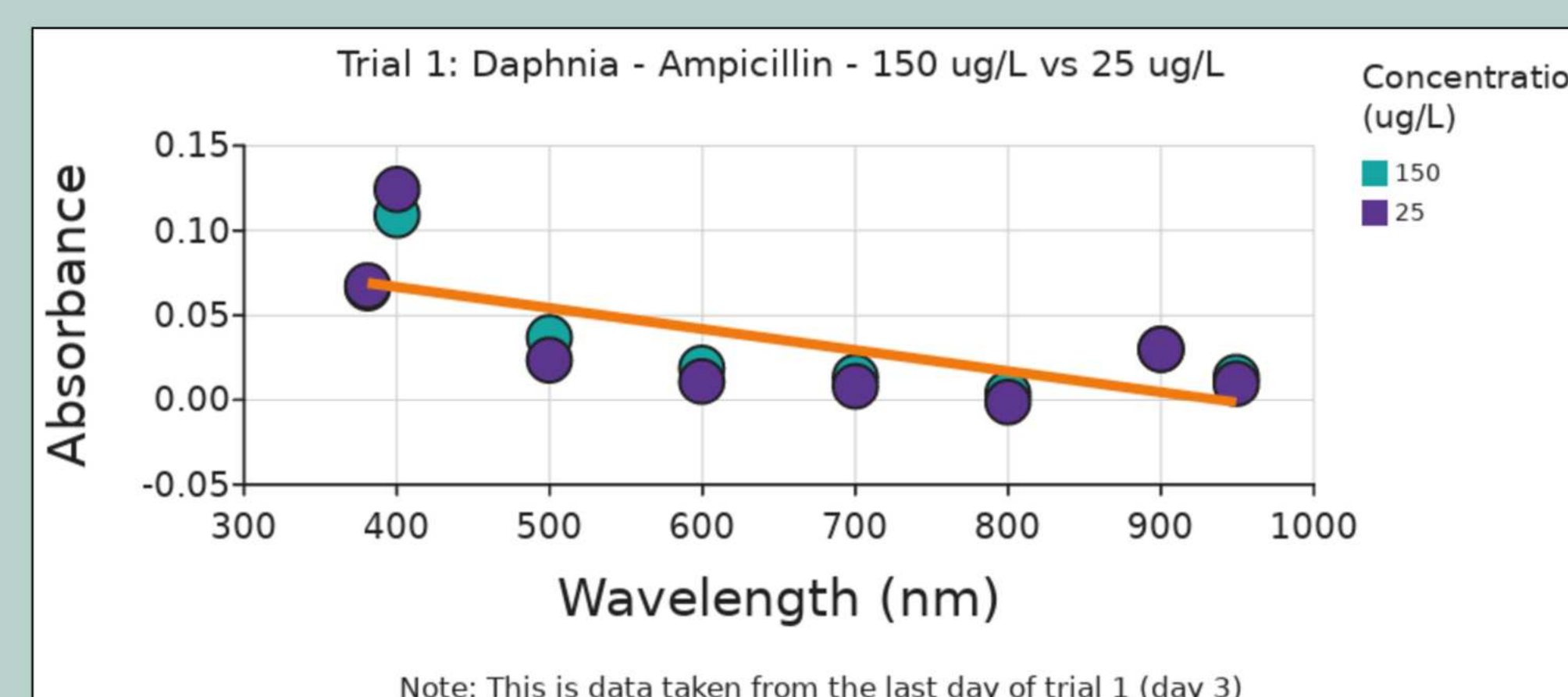


Figure 4: The effect of the presence of varying concentrations of Ampicillin on the absorbance of a solution containing *Daphnia magna*

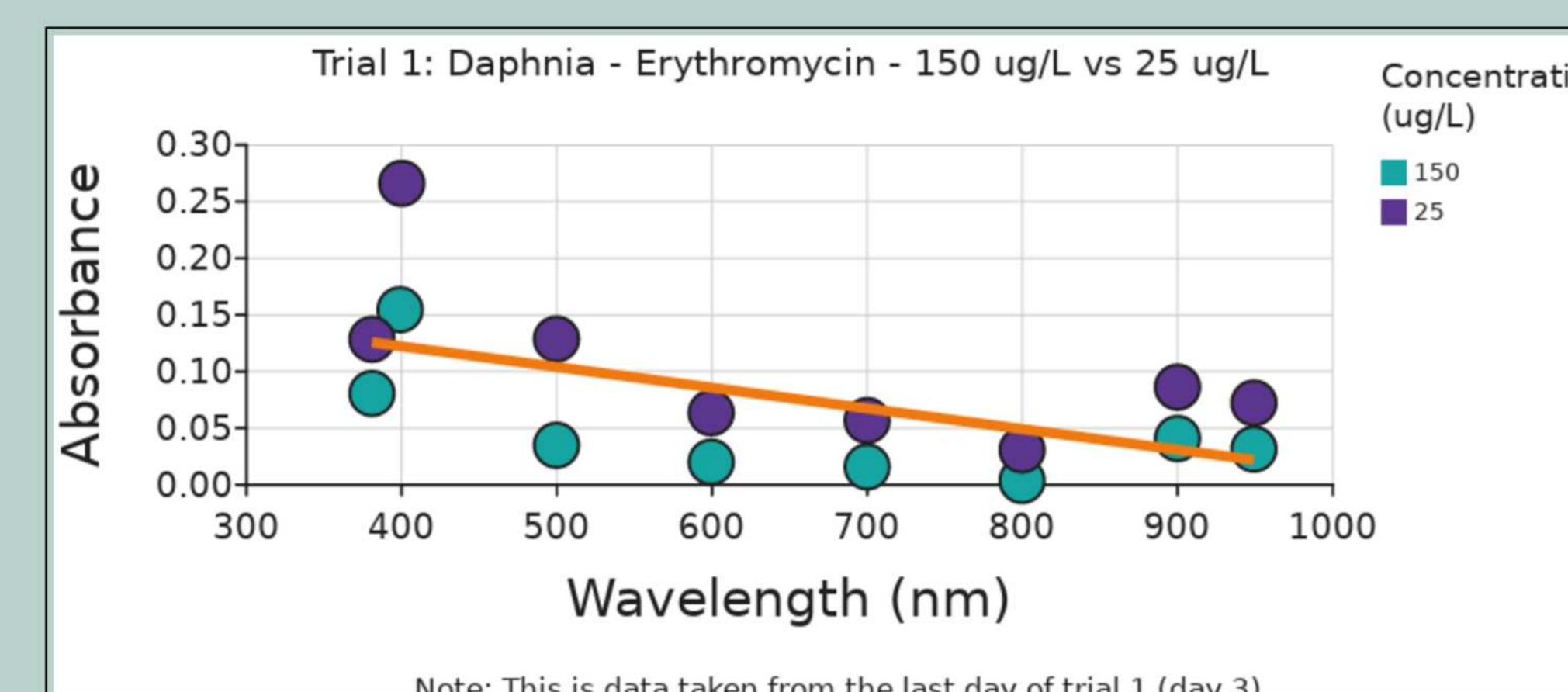


Figure 5: The effect of the presence of varying concentrations of Erythromycin on the absorbance of a solution containing *Daphnia magna*

Trial 1 of Ampicillin with <i>D. magna</i> :				
Effect	Degrees of Freedom	F-Statistic	P-value	Interpretation of P
Concentration (ug/L) (Z)	1	0.027	0.87	Means: no evidence that the groups might be different
Wavelength (nm) (X)	1	12	0.01	Means: very strong confidence that there is relationship
Interaction (X*Z)	1	0.025	0.88	Means: no evidence that the effect of X depends on Z
Error or Residual	12			

Table 1: ANCOVA Statistical Test Results for Figure 4

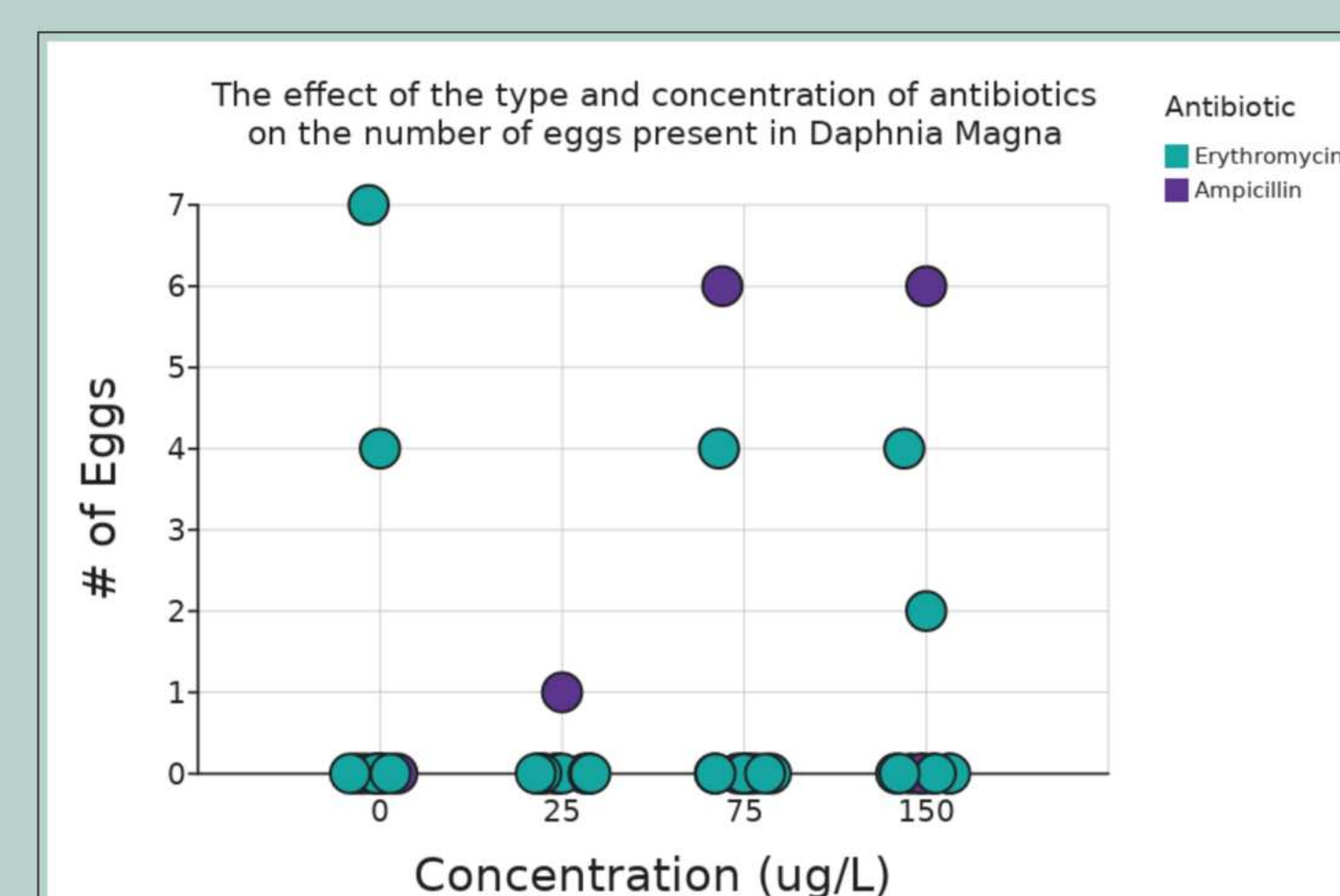


Figure 6: Shows the antibiotics, their variation and how many eggs were recorded for each

Trial 1 of Erythromycin with <i>D. magna</i>				
Effect	Degrees of Freedom	F-Statistic	P-value	Interpretation of P
Concentration (ug/L) (Z)	1	4.9	0.05	Means: strong confidence that the groups are different
Wavelength (nm) (X)	1	8.7	0.01	Means: very strong confidence that there is relationship
Interaction (X*Z)	1	0.48	0.50	Means: little to no indication that the effect of X depends on Z
Error or Residual	12			

Table 2: ANCOVA Statistical Test Results for Figure 5

Degrees of Freedom (df)	U-Statistic	Z-Statistic	P-value	Interpretation of P
46	219	0.071	0.93	A P-value of 0.93 means no evidence that the groups might be different.

Table 3: Mann-Whitney U Test Results for Figure 6

Conclusion

Algae Absorption Analysis:

An ANCOVA statistical test was used to analyze the data for absorbance of the algae using a spectrophotometer (see Figure 4, Figure 5, Table 1, and Table 2). For Ampicillin, the F-statistic was found to be 0.025 and with a p-value of 0.88. Since the F-statistic was less than the p-value, there is no evidence that the absorbance is impacted by the antibiotics.

The same statistical test was used to analyze the data for absorbance of the algae with Erythromycin. The F-statistic was found to be 0.48 with a p-value of 0.50. Once again, the F-statistic was less than the p-value, meaning there was little to no indication that the absorbance was impacted by the antibiotic.

Egg Count Analysis:

Our data does not show a statistically significant difference in the mean egg counts of the *Daphnia magna* when the concentrations and type of the antibiotics are changed. This can be seen by our P value of 0.93, which is much higher than standard significance levels of 0.05 and 0.1, indicating that our data is not statistically significant.

Based on the compiled graphs and the analysis of the data, it can be concluded that the presence of algae did not impact the antibiotics the way it was hypothesized. Additionally, the *D. magna* was not significantly impacted by the presence of the antibiotics in the water. Therefore, it is evident that the use of algae is not a viable method to reduce the amount of antibiotic residues in water.

References

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